

TITLE:

SENTINEL ONE

One Guardian. Every Digital Threat.

Hybrid AI Detection Engine for Identity Trust,
Protection & Safe Digital Banking

Team Name:

Cyber Sentinels

Team Members:

- Harshini R
- Abinaya S

Footer:

Hackathon Project | Cybersecurity & AI



Proposed Solution:

SENTINEL ONE

A **Hybrid AI Detection Engine** that protects users before fraud happens.

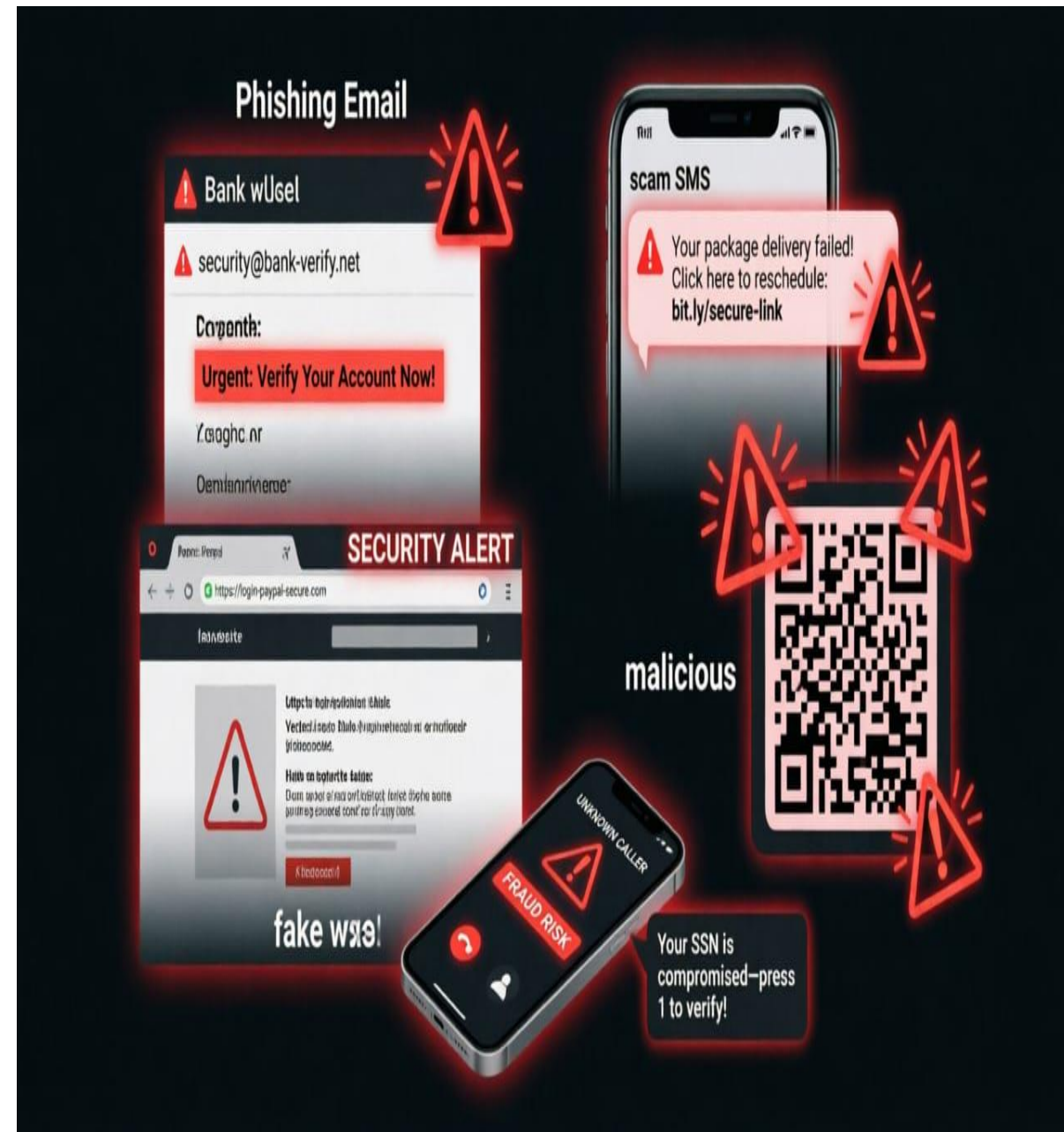
It proactively detects:

-  Fraud Emails
-  Scam SMS
-  Fake Banking Websites
-  Malicious QR Codes
-  Unknown Scam Calls

Key Benefit

Detect → Verify → Alert → Protect

Instead of reacting after fraud, SENTINEL ONE prevents fraud before users click, scan, answer, or pay.



Privacy-First Innovation & Workflow:

Privacy-First Innovation:

Our solution respects user privacy.

Privacy Features:

- ✓ Only incoming calls from unknown or unsaved numbers are analyzed.
- ✓ Saved contacts are never monitored.
- ✓ Analysis happens completely on-device.
- ✓ Conversations are never stored or uploaded.

Workflow

Email / SMS / Unknown Call / QR Code / Website



Hybrid AI Detection Engine



Threat Analysis



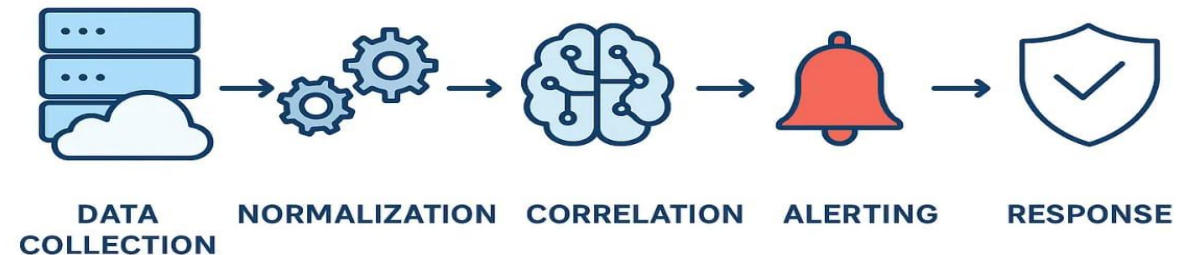
Identity Verification



User Alert



SIEM



Detection Results:

● **SAFE:**

Status

Verified Source

Reason

Trusted sender or website

Recommended Action

Safe to continue

● **SUSPICIOUS:**

Status

Potential Risk

Reason

Unverified source or unusual activity

Recommended Action

Verify before proceeding

● **FRAUD DETECTED:**

Status

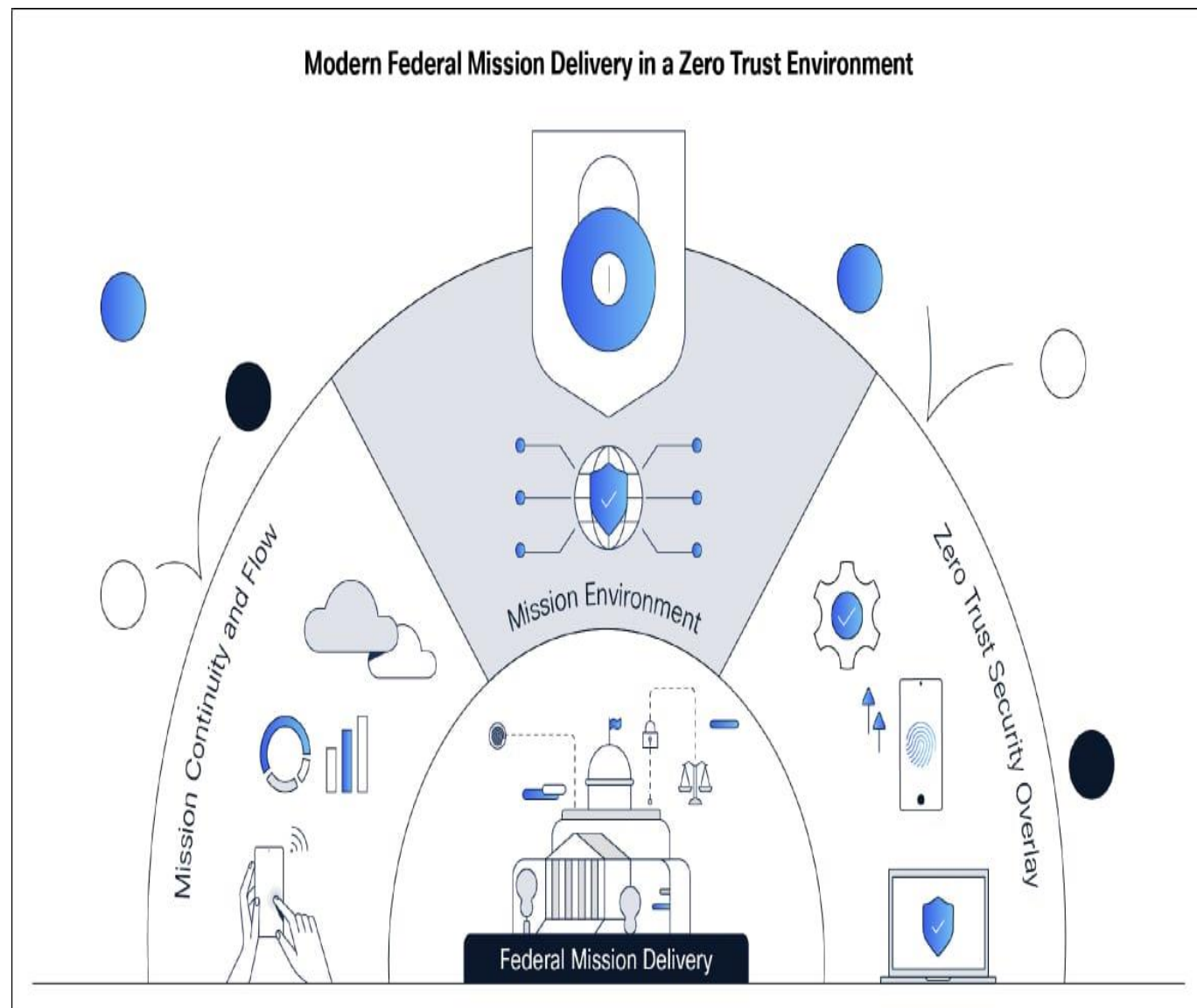
High Risk

Reason

Known phishing or scam indicators

Recommended Action

Block immediately and report



Technologies Used:

Hybrid AI Detection Engine:

Our system combines multiple AI technologies instead of relying on a single AI model.

Technologies:

- Artificial Intelligence
- Machine Learning
- Natural Language Processing (NLP)
- Computer Vision
- Optical Character Recognition (OCR)
- Speech Keyword Detection
- URL Reputation Analysis
- Phishing Detection Engine
- Cloud Threat Intelligence
- Explainable AI

Technology Mapping:

NLP → Email & SMS Analysis

Computer Vision + OCR → QR Verification

URL Analysis → Website Verification

Speech Analysis → Unknown Call Detection

AI Banking Solutions by Domain



Retail Banking

- ✓ Self-serve support and smart servicing
- ✓ Personalisation and next best action
- ✓ Explainable credit decisions



Corporate Banking

- ✓ Faster onboarding and document checks
- ✓ RM copilots and treasury insights
- ✓ Traceable, reviewable recommendations



Investment Banking

- ✓ Research summaries and deal prep
- ✓ First-pass drafting for internal notes
- ✓ Tight access control and audit trail

Competitive Advantage:

Why SENTINEL ONE?

Traditional Security

Password Protection

OTP Verification

Manual Verification

Reactive Security

No QR Protection

Limited Website Detection

Limited Email Security

Limited SMS Security

No Scam Call Detection

Limited AI Explanation

SENTINEL ONE

Hybrid AI Detection

Privacy-First Design

Multi-Channel Detection

Proactive Prevention

QR Code Detection

Website Protection

Email Protection

SMS Protection

Scam Call Detection

Explainable AI

Competitive Edge : One intelligent platform protecting every digital banking channel

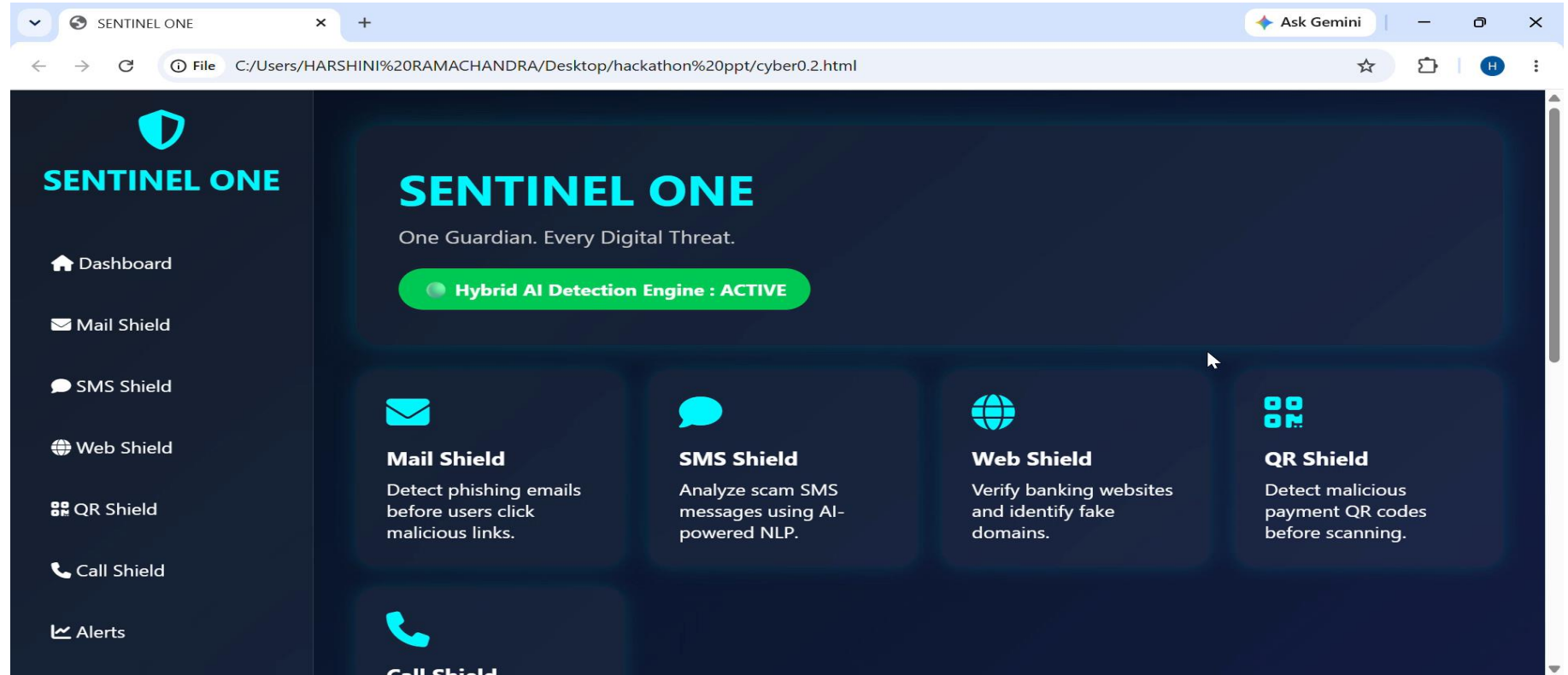


Prototype:

- **Prototype Screens**
- Dashboard
- ↓
- Unknown Call Detection
- ↓
- Email Analysis
- ↓
- SMS Analysis
- ↓
- QR Verification
- ↓
- Website Verification
- ↓
- Final Security Alert Dashboard

Dashboard Features

- Real-time threat score
- AI recommendations
- Risk level indicator
- Explainable alerts
- Security history



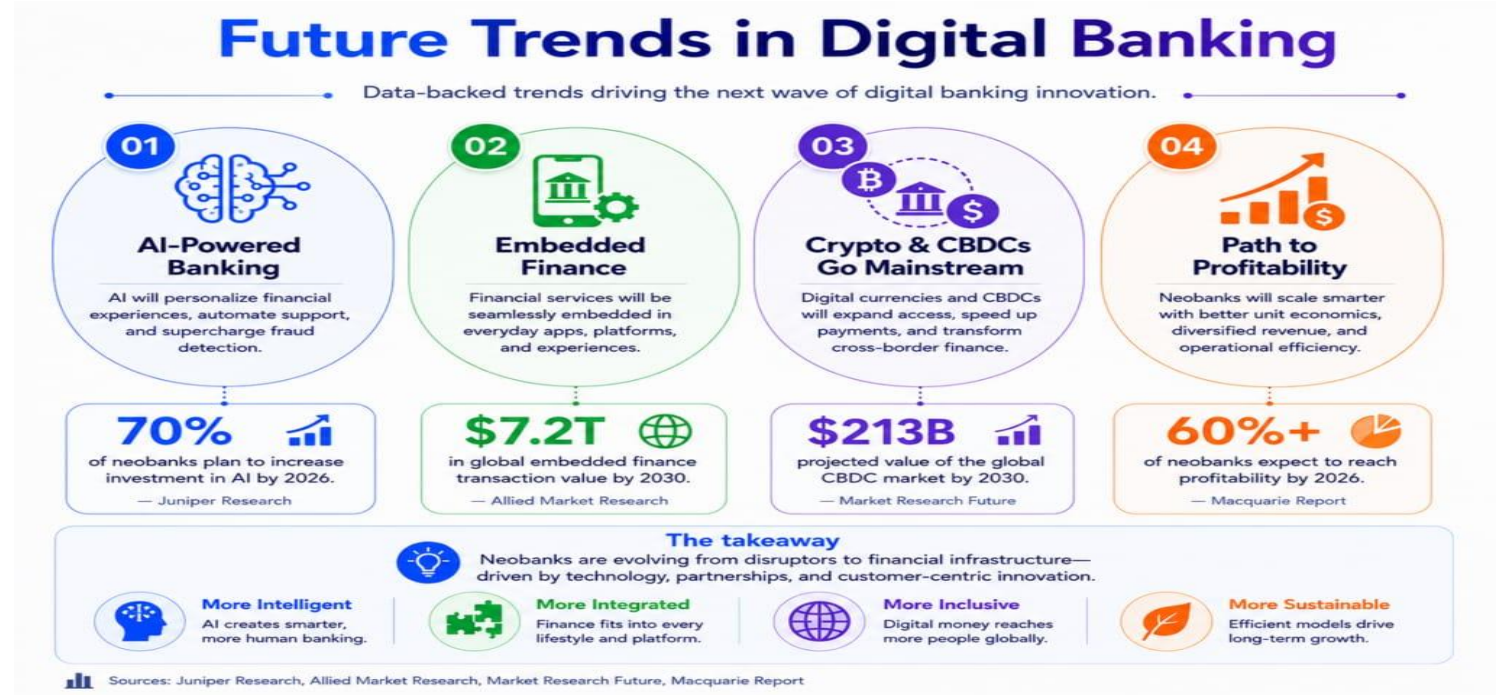
Target Users & Future Scope:

Target Users

-  Banks
-  FinTech Companies
-  UPI Users
-  Digital Banking Customers
-  Senior Citizens
-  Government Financial Services

Future Scope

- Regional Language Scam Detection
- Voice Biometrics
- Bank API Integration
- Dark Web Identity Monitoring
- Cross-Platform Protection
- Real-Time Fraud Intelligence



Conclusion:

SENTINEL ONE

One Guardian. Every Digital Threat.

Conclusion:

SENTINEL ONE is a privacy-first intelligent digital guardian that protects users across emails, SMS, QR codes, websites, and unknown phone calls before fraud happens.

By combining the **Hybrid AI Detection Engine**, Explainable AI, and proactive security, it builds trust and makes digital banking safer for everyone.

